

MAINCRAFTS TECHNOLOGY

CYBERSECURITY THREAT INTELLIGENCE REPORT

Submitted By:

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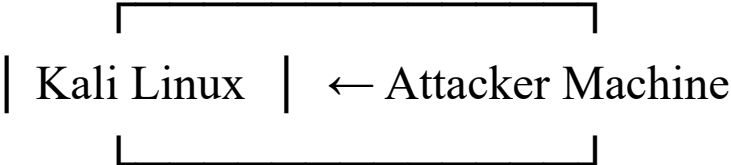
TASK-2: Setting Up a Personal Cybersecurity Lab Environment

Environment: Windows 11 Host + Kali Linux VM + OWASP
Juice Shop

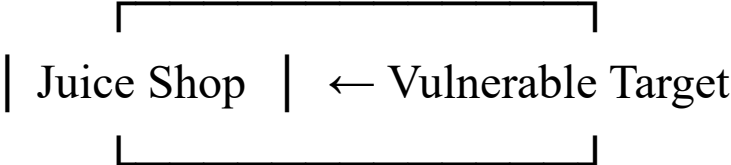
Windows Host Machine



VirtualBox



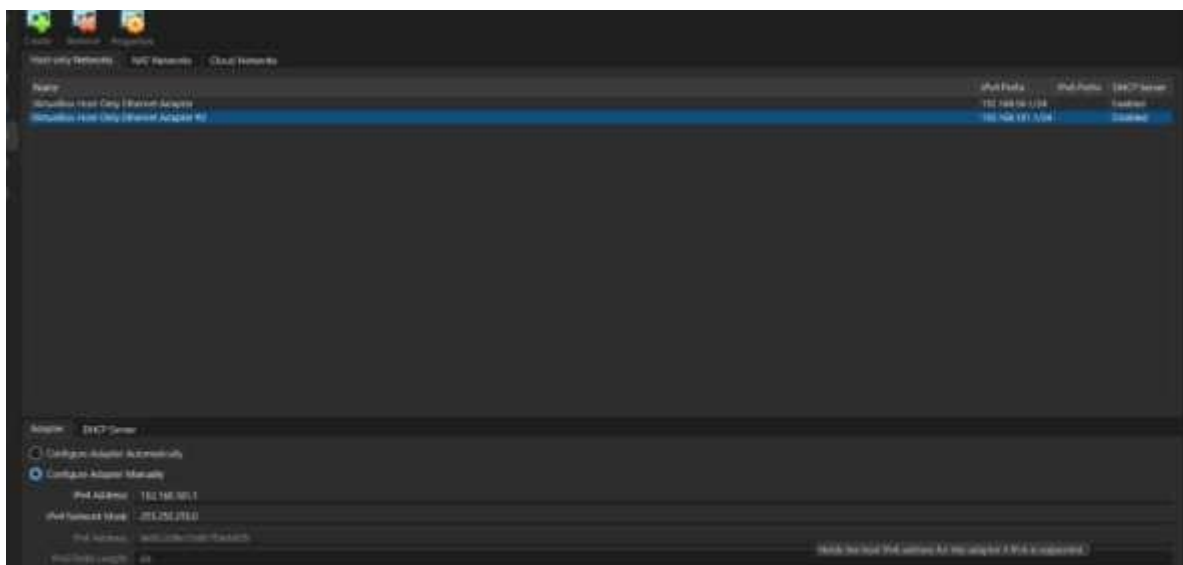
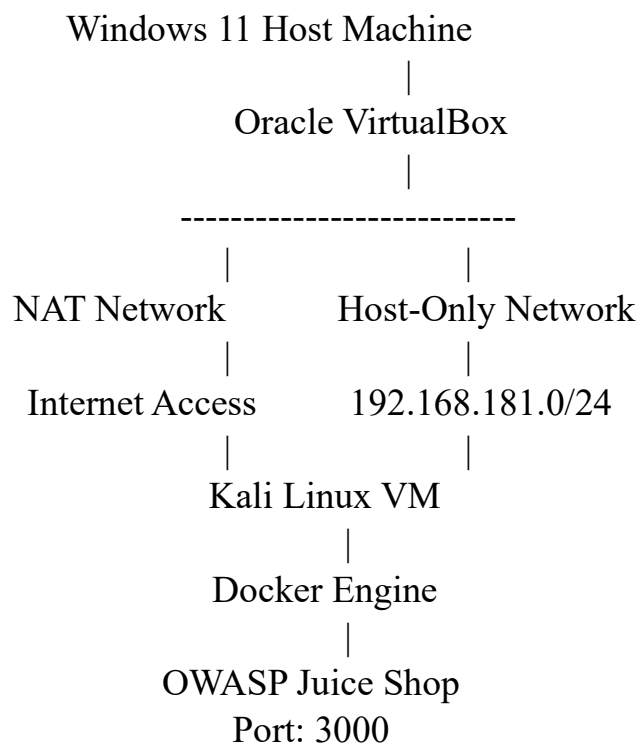
Host-Only Network



1. Objective

The objective of this task was to establish a secure and isolated cybersecurity laboratory environment for conducting penetration testing, network monitoring, and security assessment activities. The lab environment was created using Oracle VirtualBox with Kali Linux acting as the attacker machine and OWASP Juice Shop deployed through Docker as the vulnerable target application.

2. Lab Architecture



3. Environment Configuration

Component	Configuration
Host OS	Windows 11
Virtualization	Oracle VirtualBox
Attacker Machine	Kali Linux
Container Engine	Docker
Vulnerable Application	OWASP Juice Shop
Network Type	NAT + Host-Only
Host-Only Subnet	192.168.181.0/24

4. Host-Only Network Configuration

A dedicated Host-Only network adapter was created inside Oracle VirtualBox to establish an isolated environment for security testing.

Configuration Details

Parameter	Value
Adapter Name	VirtualBox Host-Only Ethernet Adapter #2
Network Address	192.168.181.1
Subnet Mask	255.255.255.0
DHCP	Disabled

5. Kali Linux Network Configuration

The Kali Linux virtual machine was configured with two network interfaces:

Interface	Purpose	IP Address
eth0	NAT	10.0.2.15
eth1	Host-Only	192.168.181.3
docker0	Docker Bridge	172.17.0.1

The network interfaces were verified using: `ip a;`

```

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:00:27:c9:10:0e brd ff:ff:ff:ff:ff:ff
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:06:05:09 brd ff:ff:ff:ff:ff:ff
    inet 192.168.101.7/24 brd 192.168.101.255 scope global dynamic noprefixroute eth1
        valid_lft 461sec preferred_lft 461sec
    inet6 fe80::a00:27ff:fe00:b509/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
4: docker0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 9e:9e:9e:9e:9e:9e brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
    inet6 fe80::dca7:f6ff:fe9e:634b/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
5: vethb1007e7d1f2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master docker0 state UP group default
    link/ether 02:70:bb:37:73:02 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet6 fe80::b070:bbff:fe37:7302/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever

```

Result

- Internet connectivity through NAT was confirmed.
- Isolated penetration testing network connectivity was established through the Host-Only adapter.
- Docker networking was successfully initialized.

6. Docker Installation Verification

Docker Engine was verified and started successfully.

Verification command:

sudo docker ps Result:

```

--> sudo docker ps
CONTAINER ID   IMAGE             COMMAND                  CREATED        STATUS        PORTS                    NAMES
4f4573b2467a   bkimminich/juice-shop   "/nodejs/bin/node /j..."   5 minutes ago   Up 3 minutes   0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp   juice

```

This confirmed that Docker daemon services were operational.

7. OWASP Juice Shop Deployment

The OWASP Juice Shop vulnerable web application was deployed using Docker.

Image Download

sudo docker pull bkimminich/juice-shop

Verification

sudo docker ps

Result: 0.0.0.0:3000 -> 3000/tcp

The vulnerable application became accessible at: <http://localhost:3000>

```
--$ sudo docker run -d -p 3000:3000 \
--name juice \
okimminich/juice-shop
4f4573b2467eeab9ff3628d2284c2149b91f2ec832e0f9af9ba32ae8a2024874
```

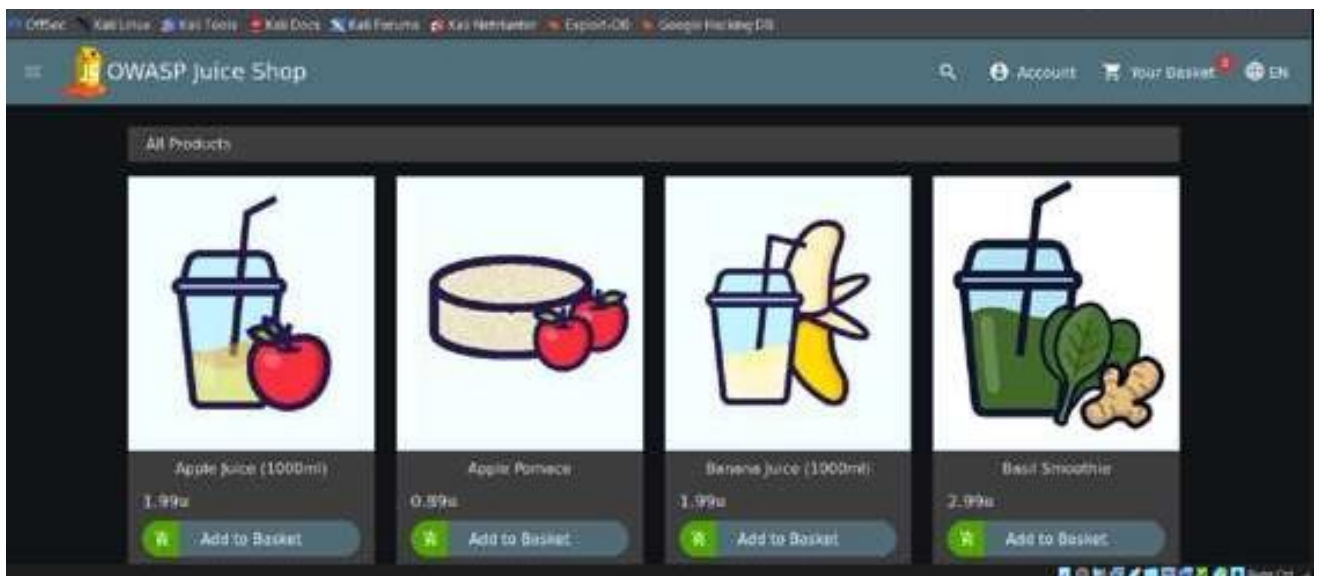
8. Browser Validation

The OWASP Juice Shop application was successfully launched inside the Kali Linux browser.

The application displayed:

- Product Catalog
- Shopping Basket
- Account Management
- Search Functionality

This validated successful deployment of the vulnerable target.



9. Network Discovery Using Nmap

Network host discovery was performed on the Host-Only network.

Command:

```
nmap -sn 192.168.181.0/24
```

Output:

Nmap scan report for 192.168.181.1

Host is up

Nmap scan report for 192.168.181.2

Host is up

Nmap scan report for 192.168.181.3

Host is up

Result:

- Three active hosts were identified on the isolated network.

```
➥$ nmap -sV 127.0.0.1 -p 3000
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-29 14:44 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00011s latency).

PORT      STATE SERVICE VERSION
3000/tcp  open  ppp?
1 service unrecognized despite returning data. If you know the service/version, please submit t
SF-Port3000-TCP:V=7.98I=7%D=6/29%Time=6A42378E%P=x86_64-pc-linux-gnu%r(Ge
SF:tRequest,2866,"HTTP/1.1\x20200\x200K\r\nAccess-Control-Allow-Origin:\x
SF:20*\r\nX-Content-Type-Options:\x20nosniff\r\nX-Frame-Options:\x20SAMEO
SF:RIGIN\r\nFeature-Policy:\x20payment\x20'self'\r\nX-Recruiting:\x20/#/jo
SF:bs\r\nAccept-Ranges:\x20bytes\r\nCache-Control:\x20public,\x20max-age=0
SF:\r\nLast-Modified:\x20Mon,\x2029\x20Jun\x202026\x2009:10:51\x20GMT\r\nE
SF:Tag:\x20W/"26af-19f12a54ea5"\r\nContent-Type:\x20text/html;\x20charse
SF:t=UTF-8\r\nContent-Length:\x209903\r\nVary:\x20Accept-Encoding\r\nDate:
SF:\x20Mon,\x2029\x20Jun\x202026\x2009:14:54\x20GMT\r\nConnection:\x20clos
SF:e\r\n\r\n!-----\n\x20\x20\x20Copyright\x20(c)\x202014-2026\x20Bjoern\x
SF:20Kimminich\x206\x20the\x20OWASP\x20Juice\x20Shop\x20contributors\x20\x20\x
SF:20\x20\x20SPDX-License-Identifier:\x20MIT\n\x20\x20→\n\n<!doctype\x2
SF:0html>\n<html\x20lang=\x20"en"\x20data-beasties-container>\n<head>\n\x20\
SF:x20<meta\x20charset=\x20"utf-8"\x20>\n\x20\x20<title>OWASP\x20Juice\x20Shop</
SF:title>\n\x20\x20<meta\x20name=\x20"description"\x20content=\x20"Probably\x20
SF:the\x20most\x20modern\x20and\x20sophisticated\x20insecure\x20web\x20app
```

10. Service Enumeration

Service detection was performed against the deployed web application.

Command:

```
nmap -sV 127.0.0.1 -p 3000
```

Result:

3000/tcp open

The Nmap service fingerprint confirmed that the web application service was successfully exposed and reachable.

```
L$ nmap -sn 192.168.181.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-29 14:45 +0530
Nmap scan report for 192.168.181.1
Host is up (0.00059s latency).
MAC Address: 0A:00:27:00:00:39 (Unknown)
Nmap scan report for 192.168.181.2
Host is up (0.00083s latency).
MAC Address: 08:00:27:4A:7B:51 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.181.3
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 2.11 seconds
```

11. Burp Suite Traffic Interception

Burp Suite Community Edition was used to intercept and analyze HTTP traffic generated by OWASP Juice Shop.

Captured request:

GET / HTTP/1.1

Host: localhost:3000

User-Agent: Mozilla/5.0

Accept: text/html

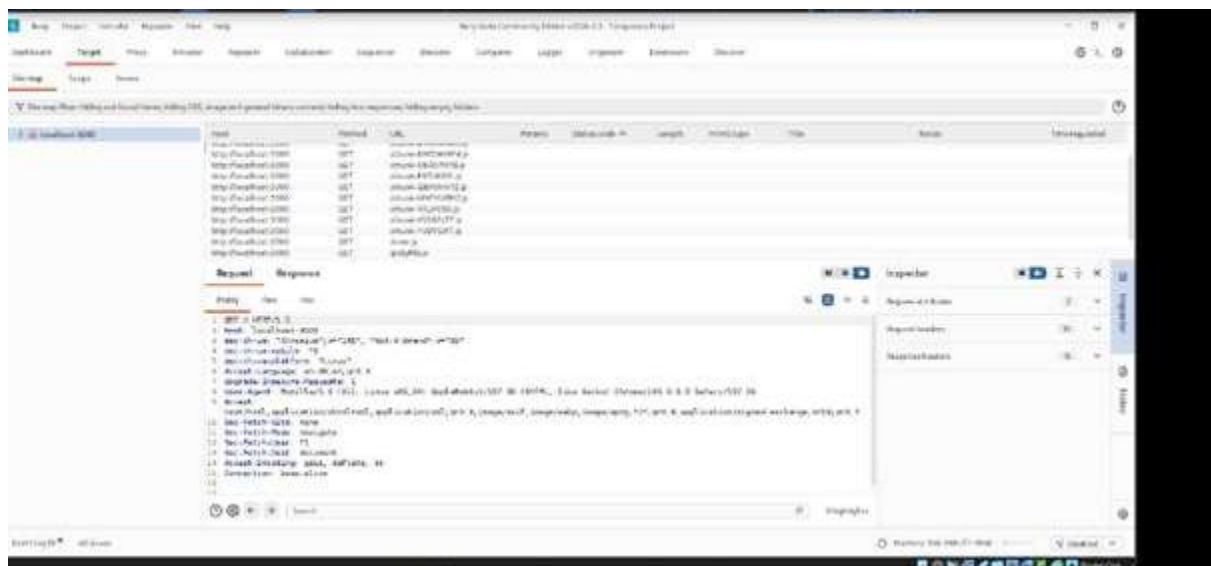
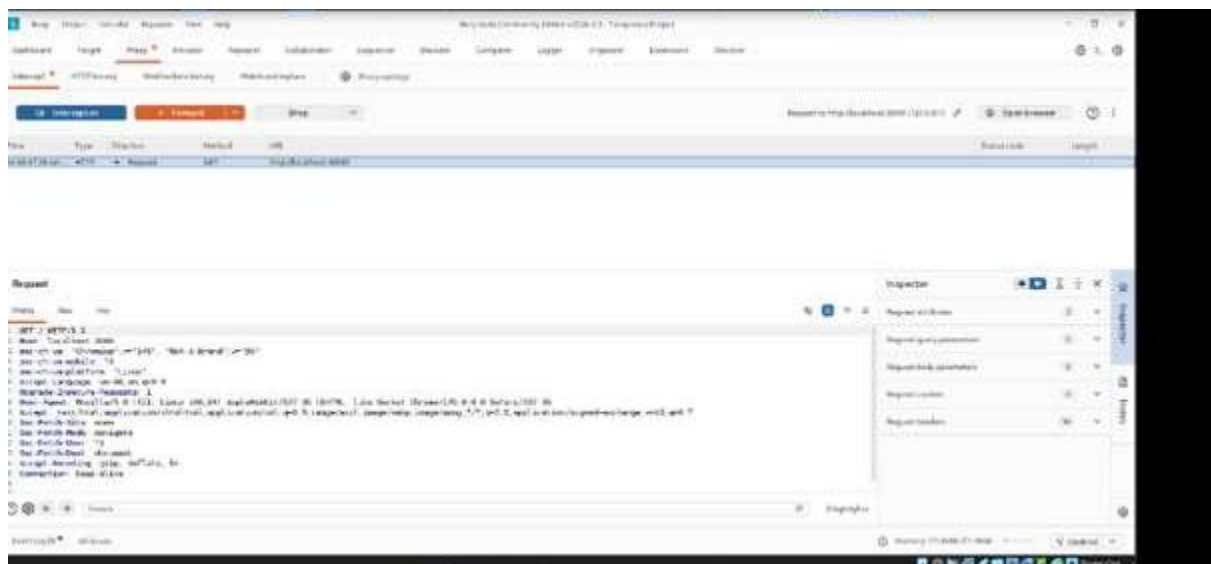
The intercepted requests appeared successfully

inside: Proxy → Intercept and were further cataloged

under:

Target → Site Map

This verified successful proxy interception and application mapping.



12. Network Packet Analysis Using Wireshark

Wireshark was used to capture network traffic generated by the OWASP Juice Shop container. Capture Filter: tcp.port==3000

Observed traffic included:

- TCP Handshakes
- ACK packets
- Retransmissions
- Application Data Transfer

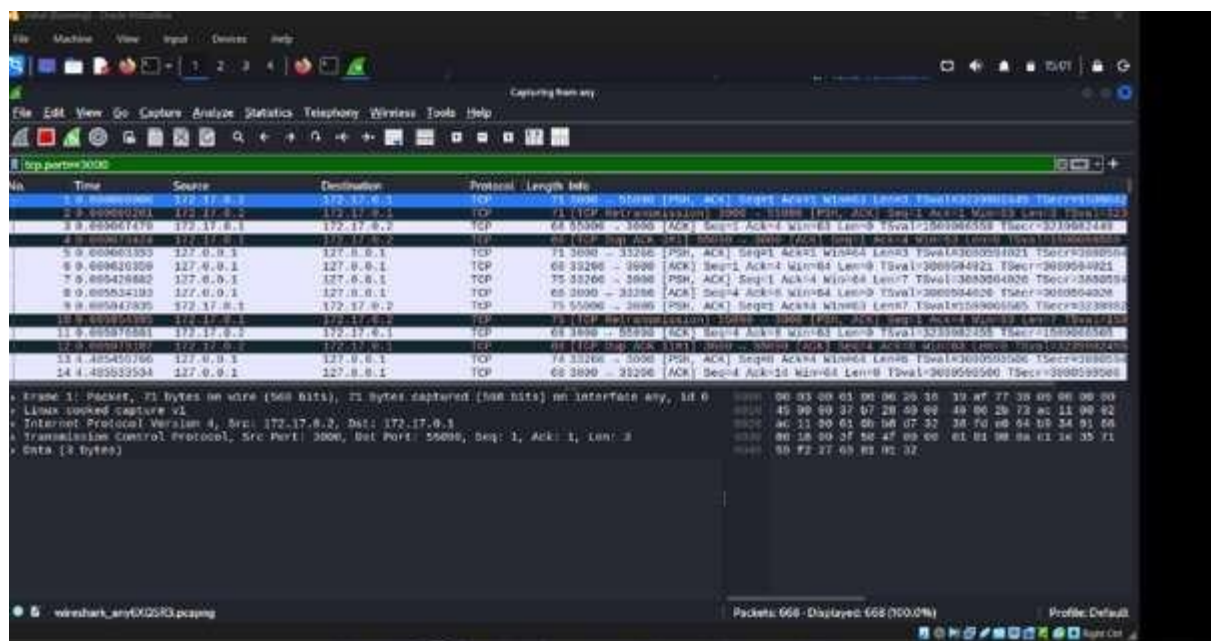
Source IP:

172.17.0.2

Destination IP:

172.17.0.1

This confirmed successful packet-level visibility into the containerized application.



13. Evidence Collected

The following evidence was collected during lab validation:

Evidence	Status
Host-Only Network Configuration	Completed
Kali Network Interfaces	Completed
Docker Verification	Completed
OWASP Juice Shop Browser Access	Completed
Nmap Host Discovery	Completed
Nmap Service Enumeration	Completed
Burp Suite Interception	Completed
Wireshark Packet Capture	Completed

14. Conclusion

A fully functional personal cybersecurity laboratory environment was successfully established using Oracle VirtualBox, Kali Linux, Docker, and OWASP Juice Shop. The environment supports penetration testing, network monitoring, traffic interception, and vulnerability assessment activities.

The lab environment provides a secure and isolated platform for conducting cybersecurity experiments without affecting production systems. Tools such as Nmap, Burp Suite, and Wireshark were successfully integrated and validated, demonstrating the effectiveness of the configured environment.